

A Compact Gesture Sensing Glove for Digital Twin of Hand Motion and Robot Teleoperation

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Abstract—Capture of hand gestures using data gloves has a wide range of applications in human–machine interaction, e.g. virtual-reality manipulation and robot teleoperation. In this article, a compact wearable sensing glove is developed using largely deformed flexible beams embedded with strain gauges. An analytical model that exactly characterizes the force-deflection behavior is established to map the local deflections of the flexible beam to the geometric constraint induced by the finger tip. In our design, ordinary carbon-fiber sheets and industry-grade strain gauges are used to make the sensing glove compact in structure and reliable for measurement. A neural network is trained using model-based simulations to run the sensing algorithm in real-time, from the strain gauges' outputs. A prototype is developed using easy-to-access materials and fabrication methods for experimental validation. The results show that the proposed sensing glove is able to capture finger motion through model-based sensing of accurate joint angle, with high compactness and sensor reliability. Demonstrations are provided to exhibit the potential of the developed sensing glove in the digital twin of hand motion and teleoperation of dexterous robotic hands.

Index Terms—Continuum mechanics, data glove, deflection sensor, motion capture, robot teleoperation.

Manuscript received 28 January 2024; revised 20 May 2024; accepted 6 June 2024. Date of publication 17 July 2024; date of current version 24 December 2024. This research was supported in part by the National Key R&D Program of China under Grant 2022YFB4701204, in part by the Natural Science Foundation of China under Grant 52022056 and Grant 52375021, and also in part by the Science and Technology Commission of Shanghai Municipality under Grant 21NL2600200. (*Corresponding author: Genliang Chen.*)

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This article has supplementary downloadable material available at <https://doi.org/10.1109/TIE.2024.3417980>, provided by the authors.

Digital Object Identifier 10.1109/TIE.2024.3417980

I. INTRODUCTION

THE use of dexterous robots is likely to become more prevalent, especially in hazardous environments, such as chemically toxic workshops, nuclear contamination and deep sea, owing to their safety, reliability, and efficiency [1], [2]. In these scenarios [3], the man-in-the-loop teleoperation is a promising technique for robot control. For the teleoperation of such complex robotic systems, the use of data gloves through human hands, rather than traditional means such as joysticks, is a more natural and flexible way for motion control [4], [5], [6], especially with the development of noninvasive and compact sensors for robot contact and motion [7].

Generally, existing data gloves consist of an array of sensors for gesture measuring, electronics, and power supply based on data acquisition and processing, as well as a holder fixed to the user's hand. Therefore, the choice of sensing units vastly influences the function and characteristics of data gloves [8]. In the past decades, a variety of sensing techniques have been adopted in the design of data gloves, including the elastomeric light guides [9], conductive liquid and liquid metal [10], [11], [12], semiconducting tape [13], [14], inertial measuring unit (IMU) [15], optical linear encoders [16], ferroelectric polymer [17], linear potentiometers [18], magnetic induction [19], and fiber Bragg gratings (FBG) [20]. However, the data gloves can be further improved in two aspects. On one hand, sensors such as IMU, FBG, and liquid metal, are complex in implementation and maintenance. On the other hand, accurate modeling of these emerging sensing methods is quite difficult for real-time hand gesture capturing.

In this article, a deflection-based sensing glove is developed for dexterous robot teleoperation based on our prior work [21]. As illustrated in Fig. 1, slender flexible beams that integrate deflection sensors at a number of discrete locations are used to capture the motion of fingers by means of coupling at the tip. To this end, a special type of bending sensor is developed by measuring the local deflections of flexible beams through industrial-grade strain gauges. Then, the finger configuration can be solved and reconstructed based on the coupling between the elastostatics of the flexible beams and the kinematics of the fingers. By using closed-form kinetostatics and integrating deflection sensors, a flexible beam can estimate its deformation under external loads in real-time [22], [23]. As shown in Fig. 1, the bending sensor is made of a slender thin carbon fiber sheet and flexible strain gauges integrated with the flexible

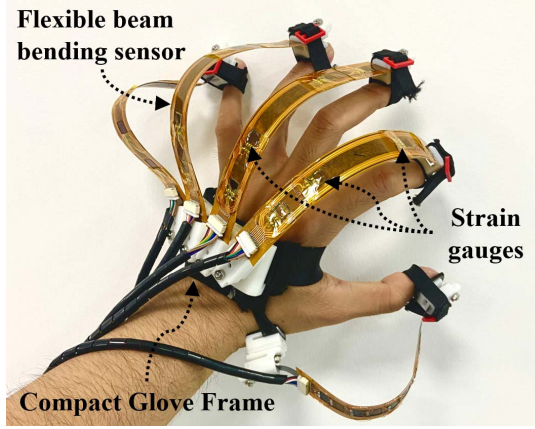


Fig. 1. Compact deflection-based sensing glove.

printed-circuit-board (FPCB). Hence, it promises a compact, lightweight, and monolithic design of bending sensors. As well, it allows easy fabrication and deployment as a wearable data glove for estimating the finger configuration. Meanwhile, a model-based algorithm is developed to capture the finger motion efficiently, and hence for the teleoperation of dexterous robotic systems in real-time.

This article is organized as follows. The architecture of the deflection-based sensing glove is proposed in Section II, where the modeling of finger kinematics and beam kinetostatics is presented as well. In Section III, the beam deformation sensing model is established. Then, the deflection-based sensing method with strain gauges is proposed and the flexible beam bending sensor is fabricated and calibrated. Then in Section IV, validation experiments are conducted to verify the feasibility of the proposed sensing glove design. A prototype of the deflection-based sensing glove is fabricated. In Section V, a data-driven algorithm for real-time finger motion capture is also developed. Demonstrations showcasing the digital twin of hand motion and teleoperation for dexterous robotic hands are exhibited. Conclusions are drawn in Section VI.

II. CONCEPTUAL DESIGN AND KINETOSTATIC MODELING

This section presents the conceptual design of the proposed sensing glove based on the large deflection of flexible beams embedded with strain gauges. A coupling model between the fingers' kinematics and the beams' kinetostatics is built up to establish the relation between the finger gesture and the beam deflection, serving as the modeling framework of the proposed sensing method for hand motion. The nomenclature for modeling is listed in Table I.

A. Conceptual Design of the Sensing Glove

The proposed sensing glove consists of two main components: the wearable connecting components and the flexible beam sensor. As shown in Fig. 2, strain gauges capable of sensing local deflections are attached to a slender flexible beam. The ends of the beam are connected to the root and tip of the fingers. Through the measurement of local bending deflections,

TABLE I
NOMENCLATURE

Symbols	Dimensions	Description
$\{S\}$	-	Spatial frame of the flexible beam and finger
$\{T\}$	-	Tip frame of the flexible beam and finger
$\mathbf{g}_{st,f}$	$\in \text{SE}(3)$	Relative finger tip pose $\{T\}$ w.r.t $\{S\}$
$\mathbf{g}_{st,b}$	$\in \text{SE}(3)$	Relative beam tip pose $\{T\}$ w.r.t $\{S\}$
$\mathbf{g}_t$	$\in \text{SE}(3)$	Target beam tip pose $\{T\}$ w.r.t $\{S\}$
φ_i	Scalar	Local deflection curvatures on the beam
α_i	Scalar	Finger joint angles
θ	$n \times 1$	Joint variables of the equivalent linkage
$\hat{\eta}_k$	$\in \text{se}(3)$	Joint twists of initial finger in standard form
$\hat{\zeta}_k$	$\in \text{se}(3)$	Joint twists of initial linkage in standard form
ζ_k	6×1	Joint twists of initial linkage in column form
ξ_i	6×1	Joint twists of current linkage in column form
$\mathbf{J}_t$	$6 \times n$	Jacobian matrix
τ	$n \times 1$	Joint force equilibrium
$\mathbf{F}_t$	6×1	External wrench
$\mathbf{K}_\theta$	$n \times n$	Joint stiffness matrix
$\mathbf{K}_J$	$n \times n$	Configuration-dependent stiffness item
y	6×1	Deviation of beam tip pose
p	4×1	Local deflection constraint
$\mathbf{S}_e$	4×1	Sensor location matrix
Φ	4×1	Sensed local curvatures

a kinetostatics-based sensing model is able to sense the large deformation of the flexible beam.

Once the finger bends, the coupled flexible beam bending sensor is simultaneously driven to deform. Since one end of the beam is connected to the finger tip, the finger and beam are coupled in a common tip pose, from which a kinematic relation of the beam and finger can be established as

$$\mathbf{g}_{st,f}(\alpha_1, \alpha_2, \alpha_3) = \mathbf{g}_t = \mathbf{g}_{st,b}(\varphi_1, \varphi_2, \varphi_3, \varphi_4) \quad (1)$$

where $\mathbf{g}_t$ is the coupling pose of the beam tip and the finger tip. $\mathbf{g}_{st,f}$ is the finger kinematics model that solves the finger tip pose through finger joint angles $\alpha_1, \alpha_2, \alpha_3$. $\mathbf{g}_{st,b}$ is the deformation sensing model of the flexible beam, which solves the beam tip pose through four discrete local deflections $\varphi_1, \varphi_2, \varphi_3, \varphi_4$ on the beam. Thus, a deflection-based finger capture conception is established, in which the finger joint angles are solved through flexible beam deflections.

B. Kinematic Modeling of Human Finger

Generally, a human finger consists of three joints, namely the metacarpophalangeal (MCP), the proximal Interphalangeal (PIP), and the distal Interphalangeal (DIP) joints [24]. In this research, all three joints (DIP, PIP, and MCP) are restricted to one degree-of-freedom (DOF) hinge, enabling finger flexion and extension. Hence, the human finger can be modeled as a three-DOF serial linkage with three revolute joints. As a

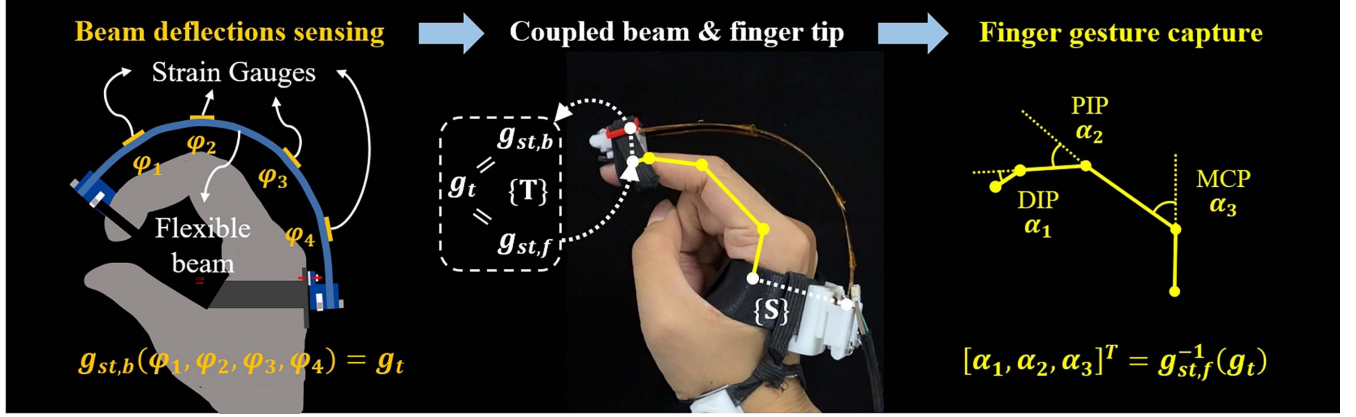


Fig. 2. Conceptual design of the deflection-based sensing glove.

result, its forward kinematics can be represented in the form of product-of-exponential (POE) formula as

$$\mathbf{g}_{st,f} = \exp(\hat{\eta}_1 \alpha_1) \exp(\hat{\eta}_2 \alpha_2) \exp(\hat{\eta}_3 \alpha_3) \mathbf{g}_{st,f0} \quad (2)$$

where $\mathbf{g}_{st,f0}$ is the initial pose of the finger tip when all joints are set to their home positions. $\hat{\eta}_1$, $\hat{\eta}_2$, and $\hat{\eta}_3$ are the twists of the revolute joints, in 3×3 matrix form of Lie algebra.

As for gesture sensing, the inverse kinematics of the finger linkage can be used to identify the joint variables in terms of the tip pose. According to the deflection-based sensing model of the flexible beam, it yields

$$[\alpha_1 \ \alpha_2 \ \alpha_3]^T = \mathbf{g}_{st,f}^{-1} \circ \mathbf{g}_{st,b}(\varphi_1, \varphi_2, \varphi_3, \varphi_4) \quad (3)$$

where $\mathbf{g}_{st,f}^{-1}$ means the inverse kinematic model of the finger, which is equivalent to that of a 3R planar open chain. The joint variables can be determined uniquely from the inverse function owing to the joints' unidirectional motion.

C. Kinetostatic Modeling of the Flexible Beams

In order to achieve the beam deformation sensing through its local deflections, kinetostatic modeling of the flexible beams is a key problem. A discretization-based method for large deflection problems of the flexible continuum, developed in our prior work [25], is introduced as the modeling framework for the characterization of the kinetostatics behavior of the slender flexible beam. Based on the principal axes decomposition theory [26], a flexible beam is discretized into a number of small segments, whose structural compliance is approximated by articulated rigid-body six-DOF linkages. By connecting the segments in serial with passive elastic joints, the flexible beams can be characterized by hyperredundant multibody systems (MBS). The advantage of this modeling approach is the approximation of continuum structural mechanics to a rigid MBS, through which the application of conventional methods for robot kinematics and statics can be used to analyze the complex continuum large deformation problems [27], [28]. In our case, the beam's thickness is significantly smaller than its width and length, therefore the shear and compression force within the beam are neglected, leaving the bending and torsion

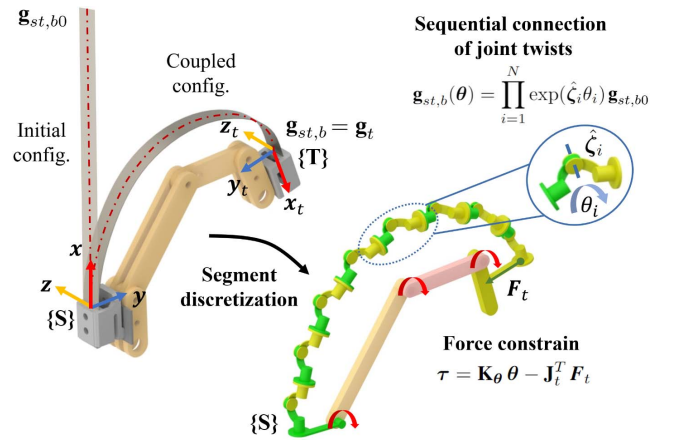


Fig. 3. Elastostatics modeling of the deformed flexible beam coupled with finger.

terms. According to the DOF of the finger joints, the effects of torsion are trivial. Thus, in our application, only bending terms are considered. In this manner, a hyperredundant MBS is constructed to model the flexible beam.

As drawn in Fig. 3, the spatial frame and the tip frame of the flexible beam are termed $\{S\}$ and $\{T\}$, respectively. Let $\theta = [\theta_1, \dots, \theta_n] \in \mathbb{R}^{n \times 1}$ be the vector of joint variables of the equivalent serial mechanism with n elastic joint. Then, the geometric constraint to the beam's tip frame can be obtained according to the forward kinematics in the form of the POE formula as

$$\mathbf{g}_{st,b}(\theta) = \exp(\hat{\zeta}_1 \theta_1) \cdots \exp(\hat{\zeta}_n \theta_n) \mathbf{g}_{st,b0} = \mathbf{g}_t \quad (4)$$

where $\mathbf{g}_{st,b} \in \text{SE}(3)$ denotes the relative pose of $\{T\}$ with respect to $\{S\}$. Here, $\mathbf{g}_{st,b0} = \mathbf{g}_{st,b}(0)$ is the initial pose, while $\mathbf{g}_t$ relates to the target one. $\hat{\zeta}_k \in \text{se}(3)$ for $k = 1, \dots, n$ are the joint twists, represented in 4×4 standard form.

Meanwhile, in the joint space, the restoring torques of the elastic joints should be balanced by the reaction force induced by the external load applied to the tip. Then, the equilibrium condition can be expressed in a concise form as

$$\tau = \mathbf{K}_\theta \theta - \mathbf{J}_t^T \mathbf{F}_t = \mathbf{0} \quad (5)$$

where $\mathbf{K}_\theta = \text{diag}(k_1, \dots, k_n) \in \mathbb{R}^{n \times n}$ represents the diagonal stiffness matrix of the elastic joints which are determined by the beam's mechanical properties. $\mathbf{F}_t \in \mathbb{R}^{6 \times 1}$ denotes the external wrench applied by finger tip at $\{\mathbf{T}\}$, which generates the forced deflection of the flexible beam. $\mathbf{J}_t \in \mathbb{R}^{6 \times n}$ is the Jacobian matrix relating the joint space to the tip frame, which can be derived as

$$(\dot{\mathbf{g}}_{st,b} \mathbf{g}_{st,b}^{-1})^\vee = \mathbf{J}_t \dot{\boldsymbol{\theta}} \Rightarrow \mathbf{J}_t = [\boldsymbol{\xi}_1, \boldsymbol{\xi}_2, \dots, \boldsymbol{\xi}_n] \quad (6)$$

where $(\cdot)^\vee$ is the 'vee' operator mapping elements of $\text{se}(3)$ from a 4×4 matrix form to the 6×1 vector one. $\boldsymbol{\xi}_i \in \mathbb{R}^{6 \times 1}$ relates to the joint twist in the current configuration of the equivalent mechanism at $\boldsymbol{\theta}$. Using the POE formula, it can then be obtained directly as

$$\boldsymbol{\xi}_i = \text{Ad} \left(\exp(\hat{\zeta}_1 \theta_1) \cdots \exp(\hat{\zeta}_{i-1} \theta_{i-1}) \right) \zeta_i, \quad i = 1, \dots, n$$

where $\text{Ad}(\cdot)$ is the adjoint operator for $\text{SE}(3)$. Refer to [29] for more details.

By combining the geometric constraint (4) at the tip and the static equilibrium (5) in the joint space, a kinetostatic model of the equivalent mechanism can be established in the form of nonlinear algebraic equations as

$$\mathbf{C}(\boldsymbol{\theta}, \mathbf{F}_t) = \begin{bmatrix} \mathbf{y} \\ \boldsymbol{\tau} \end{bmatrix} = \begin{bmatrix} (\ln(\mathbf{g}_{st,b} \mathbf{g}_t^{-1}))^\vee \\ \mathbf{K}_\theta \boldsymbol{\theta} - \mathbf{J}_t^T \mathbf{F}_t \end{bmatrix} = \mathbf{0} \quad (7)$$

where $\mathbf{y} \in \mathbb{R}^{6 \times 1}$ represents the pose deviation of the beam tip frame from the target one $\mathbf{g}_t$, while $\boldsymbol{\tau} \in \mathbb{R}^{n \times 1}$ relates to the deviation of torques in the elastic joints to balance the external load applied at the tip.

This kinetostatic model relates to a set of nonlinear algebraic equations with $6 + n$ equations and $n + 6$ unknowns $\boldsymbol{\theta}$, $\mathbf{F}_t$. Since both $\mathbf{g}_t$ and $\mathbf{F}_t$ degenerate to the planar configuration, the numbers of the valid equations and the independent unknowns are reduced to $n + 3$. It implies that the kinetostatics model is deterministic and can be solved efficiently through the classical Newton–Raphson method.

From (7), a closed-form solution of the system Jacobian matrix with respect to $\boldsymbol{\theta}$ and $\mathbf{F}_t$ can be derived as

$$\nabla \mathbf{C} = \begin{bmatrix} \frac{\partial \mathbf{y}}{\partial \boldsymbol{\theta}} & \frac{\partial \mathbf{y}}{\partial \mathbf{F}_t} \\ \frac{\partial \boldsymbol{\tau}}{\partial \boldsymbol{\theta}} & \frac{\partial \boldsymbol{\tau}}{\partial \mathbf{F}_t} \end{bmatrix} = \begin{bmatrix} \mathbf{J}_t & \mathbf{0} \\ \mathbf{K}_\theta - \mathbf{K}_J & -\mathbf{J}_t^T \end{bmatrix} \quad (8)$$

where $\mathbf{K}_J \in \mathbb{R}^{n \times n}$ is a configuration-dependent stiffness item. Please refer to [25] for more details. The nonlinear algebraic equation can then be solved in an iterative way, by following the update theme as:

$$\begin{bmatrix} \boldsymbol{\theta} \\ \mathbf{F}_t \end{bmatrix}^{(k+1)} = \begin{bmatrix} \boldsymbol{\theta} \\ \mathbf{F}_t \end{bmatrix}^{(k)} - (\nabla \mathbf{C}^{(k)})^{-1} \mathbf{C}^{(k)} \quad (9)$$

where the superscript ' k ' corresponds to the variables obtained in the k th iterative step.

III. SENSING METHOD AND MODELING FRAMEWORK

Based on the kinetostatic model derived in Section II-C, this section presents the development of the sensing model through

the deformed flexible beam. An approach for local deflection measuring is also provided, with the calibration for identifying the properties of the flexible strain gauges.

A. Sensing Model of the Flexible Beam Deformation

In the scenario of gesture capturing, the coupled tip pose $\mathbf{g}_t$ is to be sensed from local deflections. Geometric constraints can then be induced to the beam's local deflections related to the strain gauges. Thus, the constraints can be represented as

$$\mathbf{p} = \mathbf{S}_e \boldsymbol{\theta} - \boldsymbol{\Phi} = \mathbf{0} \quad \text{with} \quad \boldsymbol{\Phi} = [\varphi_1, \varphi_2, \varphi_3, \varphi_4]^T \quad (10)$$

where $\varphi_1, \varphi_2, \varphi_3, \varphi_4$ are the deflection angles measured by the strain gauges. $\mathbf{S}_e \in \mathbb{R}^{4 \times n}$ is a constant matrix that maps the beam's discretized joint angles to the deflections of sensor-related segments. The rows of $\mathbf{S}_e$ are given as

$$\mathbf{S}_j = [0, \dots, 0, 1, \dots, 1, 0, \dots, 0], \quad j = 1, \dots, 4$$

whose $n_{j,1} \sim n_{j,2}$ elements are ones and the others are zeros, where $n_{j,1}$ and $n_{j,2}$ are the numbers of joints in front of the sensor's proximal and distal ends, respectively.

Composing the geometric constraint to the local deflection (10) and the static equilibrium (5), the sensing model of the flexible beam can be obtained as

$$\mathbf{S}(\boldsymbol{\theta}, \mathbf{F}_t) = \begin{bmatrix} \mathbf{p} \\ \boldsymbol{\tau} \end{bmatrix} = \begin{bmatrix} \mathbf{S}_e \boldsymbol{\theta} - \boldsymbol{\Phi} \\ \mathbf{K}_\theta \boldsymbol{\theta} - \mathbf{J}_t^T \mathbf{F}_t \end{bmatrix} = \mathbf{0}. \quad (11)$$

Similarly, the system Jacobian matrix of the sensing model can be derived as

$$\nabla \mathbf{S} = \begin{bmatrix} \frac{\partial \mathbf{p}}{\partial \boldsymbol{\theta}} & \frac{\partial \mathbf{p}}{\partial \mathbf{F}_t} \\ \frac{\partial \boldsymbol{\tau}}{\partial \boldsymbol{\theta}} & \frac{\partial \boldsymbol{\tau}}{\partial \mathbf{F}_t} \end{bmatrix} = \begin{bmatrix} \mathbf{S}_e & \mathbf{0} \\ \mathbf{K}_\theta - \mathbf{K}_J & -\mathbf{J}_t^T \end{bmatrix}. \quad (12)$$

Since $\mathbf{F}_t \in \mathbb{R}^{3 \times 1}$ relates to a planar wrench, the number of unknowns equals $n + 3$. As $\boldsymbol{\tau} = \mathbf{0}$ provides n equations, $\mathbf{p} = \mathbf{0}$ should introduce at least three extra constraints to solve the sensing model. In other words, at least three strain gauges are required to measure the beam's local deflection at discrete positions. To increase computational robustness and accuracy, four strain gauges are used for each finger in the developed sensing glove as illustrated in Fig. 2.

B. Measuring System of the Beam's Local Deflections

As illustrated in Fig. 4, four customized strain sensors are deployed on a carbon fiber flexible beam at discrete locations to measure the local deflections of the flexible beam. A full-bridge, 2×2 grid design is adopted to reduce the influence of measuring noise and temperature drift of the strain gauges. The flexible printed circuit board (FPCB) is used to integrate connection wires of the strain gauges, which keeps the measuring system compact. Such integration also prevents extra interference caused by the measuring system from affecting the beam's kinetostatic model. Sampling circuits including low-pass filter, amplifier, A/D converter, Arduino and linear power units are also integrated into a wearable and compact scale.

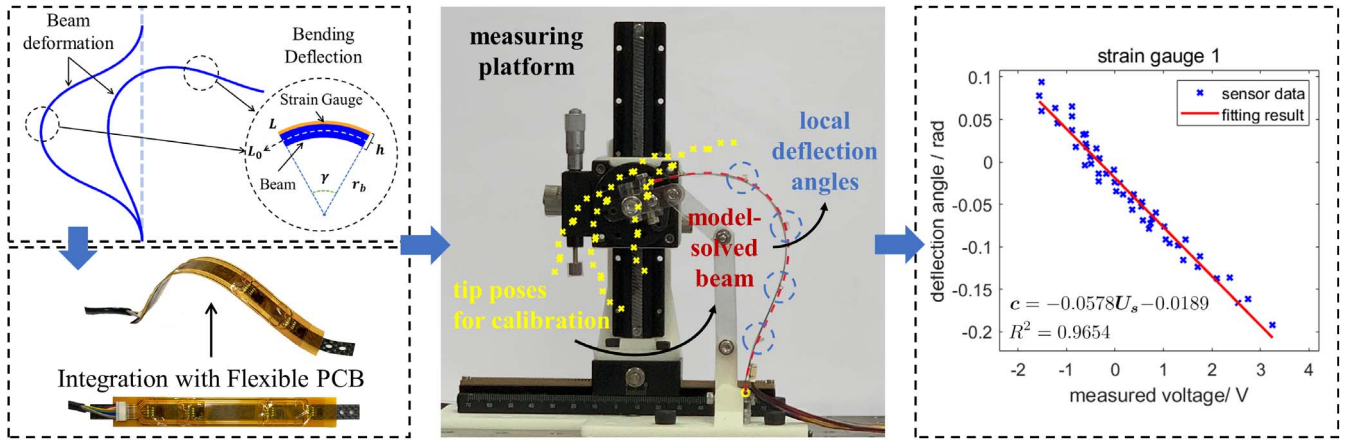


Fig. 4. Deflection sensing method using strain gauges: concept, design, and calibration.

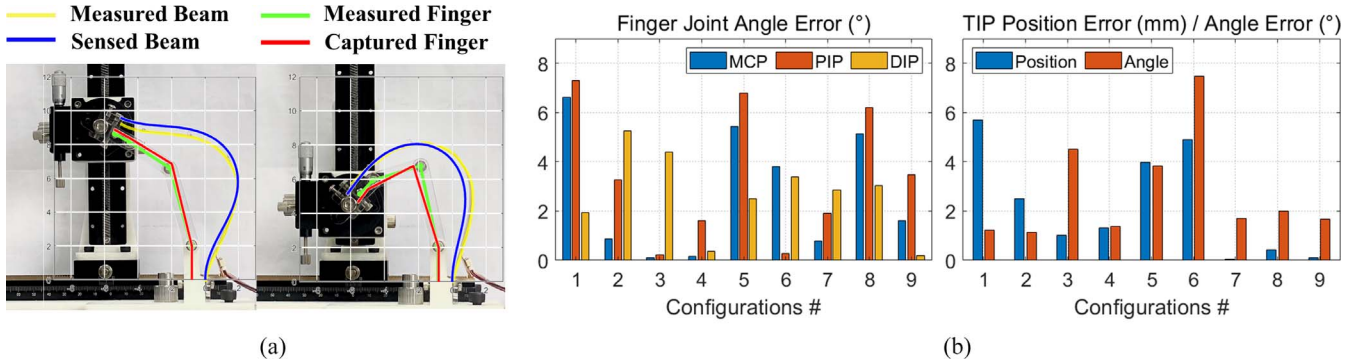


Fig. 5. Validation of finger gesture capturing. (a) Visualization of the sensed/measured finger. (b) Sensing errors of joint angles and tip pose.

According to the characteristics of the full-bridge strain gauge, a linear relation between the deflection curvatures c and the voltage U_s of strain gauges can be expressed as

$$c = \frac{8l_0}{Kh_0U_0}U_s = \lambda U_s \quad (13)$$

where U_0 and U_s denote the supply and output voltages of the strain gauge. h_0 and l_0 relate to the beam's thickness and the effective length of the gauges. $K = (1 + 2\mu) + (1 - 2\mu)B$ is the sensitivity ratio, with Poisson's ratio μ and Bridgman's constant B of the wire material. The above parameters can be together considered as a single linear parameter λ , which can be determined in application through calibration.

C. Calibration of the Deformed Flexible Beam

The linear parameter λ of each strain gauge on the flexible beam can be obtained through linear fitting. As depicted in Fig. 4, based on common gestures achievable by the human finger, 50 configurations of the flexible beam are selected and then set by a three-DOF measuring platform. The datum of local bending curvatures under those calibration configurations is solved from the beam's kinestatics (7), while voltage outputs of strain gauges are recorded in the meantime. As the fitting example demonstrated in Fig. 4, the linear relationship is

well satisfied, and each strain sensor's curvature-voltage linear formulas can be established.

IV. VALIDATION EXPERIMENTS AND PROTOTYPING

To validate the estimation accuracy of the proposed method, sensing experiments of the beam's tip pose are conducted on a three-joint skeleton linkage. Meanwhile, a wearable prototype of a five-finger sensing glove is fabricated via easy-to-access materials and methods. In addition, a program with a graphical interface is developed to integrate the acquiring, calculation, and visualization modules.

A. Validation of the Gesture Capturing Method

To validate the effectiveness of the proposed gesture-capturing method, a skeleton linkage mimicking the finger motion is fabricated and installed on the experimental setup as shown in Fig. 5. According to the index finger size of one of the authors, a 180-mm-long carbon-fiber slender sheet, with a $10\text{ mm} \times 0.2\text{ mm}$ cross-section, is used as the flexible beam to capture the finger motion. As indicated above, four strain gauges are deployed on the flexible beam to acquire sufficient information on local deflection. All these strain gauges have been calibrated in advance, with the linear parameters given in

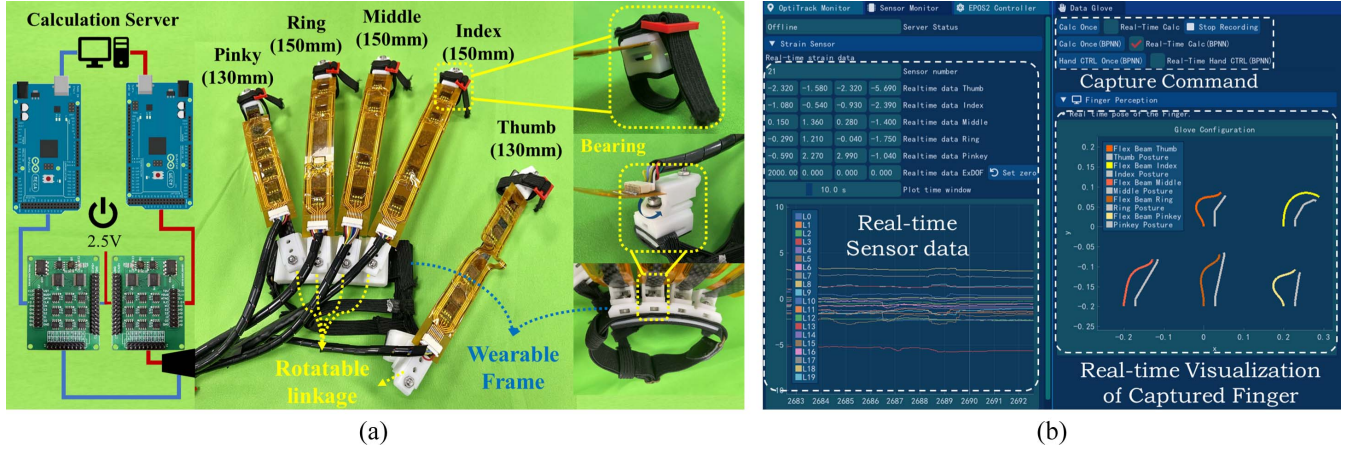


Fig. 6. Sensing glove prototype. (a) System design. (b) C++-based real-time capture program with graphical interface.

TABLE II
FITTED CURVATURE - TENSION FORMULAS

Linear param.	Strain Gauges #			
	1	2	3	4
Slope	-0.0578	-0.0536	-0.0467	-0.0372
Intercept	-0.0189	0.0289	0.0304	0.0674
R^2	0.9654	0.9618	0.9699	0.9712

Table II. The sensor outputs are collected through the signal sampling circuit, as introduced in Section III-C.

In this experiment, nine configurations are prescribed for the finger's skeleton linkage, and the corresponding positions and orientations of the finger tips are recorded as well. Meanwhile, according to (13), the local deflection of the strain gauges can be determined in terms of the acquired output voltages. Then, the pose of the finger tip can be estimated by means of solving the sensing model (11) of the flexible beam. Further, the finger gesture can be determined through the inverse kinematics (3). Two examples of gesture capturing are provided in Fig. 7, with a comparison of the measured intuitively. From the results, it can be seen that both the deformed flexible beam and the finger gesture that is rebuilt from the proposed sensing method, well fit the actual situation. The results of all the nine tested configurations are illustrated in Fig. 7. From the results, it is known that the mean position and orientation estimation errors of the tip pose are 2.2 mm and 2.8° , respectively. The mean estimation error of the finger joints is 3.0° . While max values of errors are: 7.3° and 5.7 mm for tip poses, and 7.5° for the joint angles. It is quite acceptable for such a simple prototype fabricated only with industrial-grade components, as factors such as misalignment and measuring errors of strain gauges are not taken into account in the sensing model.

B. Prototyping of the Sensing Glove

After validating the feasibility and accuracy of flexible beam bending sensors, a sensing glove prototype is fabricated with

a customized wearable frame, flexible beam fixation mechanism, and signal sampling circuit, as shown in Fig. 6. Five flexible beams embedded with strain gauges are connected to the signal sampling circuit. A total of 20 measured voltages are transmitted by Arduino Mega 2560 to the serial port of the calculation server. In the software, the kinetostatic calculation is implemented in a C++ environment. By adopting the SoftBUS technique, signal sampling, capture visualization, and robot communication are also integrated. Based on the above framework, a capture program with a graphical user interface is then created as illustrated in Fig. 6.

In terms of real-time performance, the capture frequency is determined by the sampling speed of the strain gauges and the calculation speed of the kinetostatic sensing model. The capture test is conducted on a laptop. The voltage sampling of 20 strain gauges achieves a frequency of over 30 Hz, while the model calculation for five fingers reaches a frequency of over 40 Hz. In total, the sensing glove can achieve a capture frequency exceeding 30 Hz, thus demonstrating its real-time capture ability. This observation also validates the low computational overhead of the deflection-based finger gesture capture method, making it easy for deployment on portable hardware.

V. APPLICATION IN ROBOT TELEOPERATION

In this section, demonstrations of real-time gesture capture are provided to exhibit the potential of the developed sensing glove in the applications of robot teleoperation. Using the back-propagation neural network (BPNN) algorithm [30], a data-driven method is developed to increase the sensing efficiency for gesture capturing and motion controlling in real-time. To verify the effectiveness of the proposed method, a digital twin of hand motion is created, and a dexterous robotic hand is teleoperated using the prototype [31].

A. Data-Driven Method for Gesture Estimation

The accuracy and efficiency of the developed sensing glove have been verified via the validation experiment. However, the Newton-Euler method imposes a certain computational burden

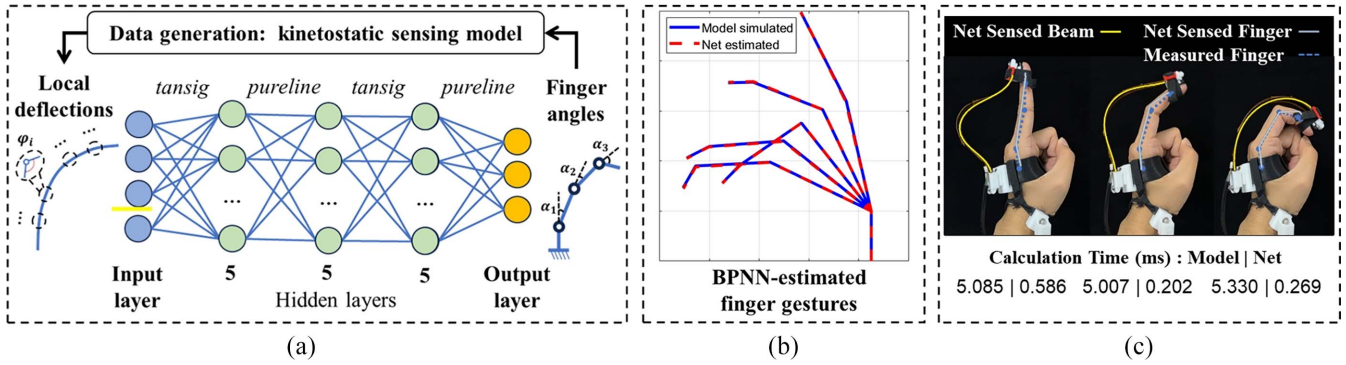


Fig. 7. Data-driven method for gesture estimation. (a) Network training using simulation results. (b) Results visualization. (c) Test via the prototype.

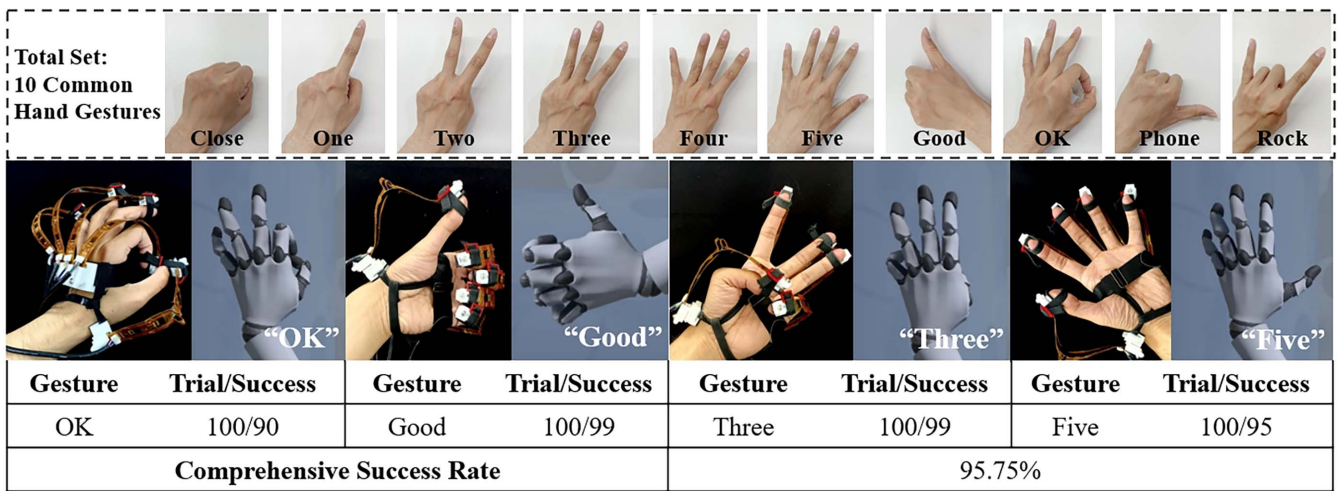


Fig. 8. Hand gesture capturing in the virtual environment unity, using the proposed sensing glove prototype.

for solving the system of nonlinear algebraic equations of the proposed sensing model. It requires a computer to implement the iterative searching algorithm for capturing the motion of fingers. Hence, the technique of neural network is employed to reduce the computational burden and simplify the deployment of the proposed sensing model on embedded systems. Since a neural network is able to solve nonlinear algebraic equations in a straightforward manner, it is suitable for the developed sensing glove to use a data-driven method for gesture capture. Further, the accuracy of the proposed sensing model has been validated, such that simulation results can be generated in a sufficient amount for training the neural network.

In the developed prototype, the BPNN algorithm which is widely used for surface fitting, is adopted to solve the sensing model of the deformed flexible beams. As drawn in Fig. 7, a three-hidden-layer BPNN with 103 parameters is established, and the *tansig* activation function is utilized for training. As indicated in the figure, the gesture of fingers can be determined directly using the neural network in terms of the beams' local deflections measured via the strain gauges. It is worth noting that individual networks are trained for different fingers due to their differences in size. For the training of each finger, 512

gestures are simulated according to the kinetostatic model of the deformed flexible beams.

The accuracy and efficiency of the data-driven method are tested through a set of gestures performed by the index finger. The results are illustrated in Fig. 7 with a comparison of those calculated via the exact sensing model. From the figure, it can be seen that the well-trained BPNN achieves notable accuracy for gesture estimation of the fingers. The networks' estimation errors for the fingers' joint variables are all less than 1° with respect to those solved from the nonlinear algebraic equations. Moreover, it only takes the BPNN about $0.5 \mu s$ to predict the finger gesture, which is 10 times faster than the exact sensing model. In addition, examples of sensing results are provided in Fig. 7 to exhibit the feasibility of the developed networks in an intuitive way.

B. Demonstration of Digital Twin and Teleoperation

To exhibit the potential applications of the proposed sensing glove, demonstrations of the digital twin of hand motion and the teleoperation of dexterous robotic hands are conducted on the developed prototype. As shown in Fig. 8, the wearable



Fig. 9. Teleoperation on a real dexterous robot hand: hand gesture reproduction “number one”, “OK” and object interaction “grasp”, “pinch”.

five-finger sensing glove is used to create a real-time digital twin of hand motion. Different gestures are performed and estimated, and then the results are used to control a robotic hand in physical environments.

In the first test, a digital twin of hand is developed in the virtual engine of unity. The motion of the virtual hand is remotely operated by a tester through the sensing glove. As shown in Fig. 8, 10 common hand gestures are selected to establish a total recognition set, with an exclusive “X” option for results with low local sensing accuracy in part of joints. Then four representative gestures “OK”, “good”, “three”, and “five” are selected as the test samples. To ensure the randomness of sampling, a deck of gesture cards has been prepared. A total of four observers have been set up to identify the gestures of the digital twin hand. In each trial, the subject randomly selects a gesture card and makes the corresponding gesture with the sensing glove. The observer will then classify the gesture in the total recognition set according to the captured gesture of the digital twin hand, so that 100 recognition trials for each observer, a total of 400 trials have been conducted. The number of successful captures of the four gestures is recorded. The comprehensive success rate is calculated according to the number of gestures successfully recognized with respect to the total number of trials.

As can be seen, the success rate of gesture capture reaches 95.75%. A small number of gesture capturing has failed because of the two following reasons. 1) Although the proposed data glove design uses stable industrial-grade components, this simple prototype was hand-soldered and glued by the authors, and in a few cases, poor contacts can cause abnormal signal fluctuations. 2) Measurement errors exist in the glove structure such that its coupling to the hand is imperfect. Both problems can be solved by switching to industrial-grade equipment for assembly and preparation. Thus, in normal working states, the proposed sensing glove can well perform gestures reconstruction in a digital twin environment. In addition, a supplementary video¹ is provided to intuitively demonstrate the performance of the sensing glove.

Moreover, a demonstration of the teleoperation of a dexterous robotic hand is also provided to exhibit the feasibility of the developed sensing glove applied in the physical environment. As illustrated in Fig. 9, in this case, the tester’s hand motion is captured through the sensing glove and then used to control the dexterous hand. Owing to the high efficiency of the data-driven method, the teleoperation of the physical robotic hand could be

implemented in real-time. From the results, it can be seen that the robotic hand is able to precisely reproduce the tester’s hand motion to perform dexterous grasping tasks. As illustrated in the figure, teleoperation is realized successfully in grasping objects of different shapes and materials. Likewise, this example is also demonstrated in the supplementary video¹ in an intuitive way.

VI. CONCLUSION

In this article, a deflection-based sensing glove is developed for capturing hand motion and teleoperating robotic hands. Slender flexible beams embedded with industrial-grade strain gauges are adopted to make the sensors compact in structure and reliable for capturing. Closed-form models are established for the kinetostatics analysis of the deformed flexible beams, which permits an accurate and efficient solution to the sensing approach. Using the sensing glove developed in this article, not only the finger gesture but also the exact joint rotations can be determined in real-time. A wearable prototype of the sensing glove is developed using easy-to-access materials and simple fabrication methods. Using the BPNN algorithm, a data-driven method is employed for the prototype to further improve the computational efficiency of the capturing model. Experiments are carried out to validate the effectiveness and efficiency of the proposed sensing glove in gesture capturing. To exhibit the potential applications of the developed prototype, demonstrations of the digital twin of hand motion and the teleoperation of dexterous robotic hands are provided. From the results, it can be stated that the proposed deflection-based sensing glove is capable of precisely capturing the motion of human hands and efficiently performing teleoperation of physical robotic hands in real-time.

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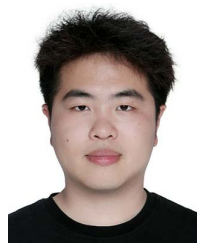
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